

Retention in Care Among PLHIV Accessing Care at Public Treatment Sites in Jamaica

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Objectives: To characterize Jamaica's HIV population based on retention status, identify factors associated with retention in care, and develop recommendations to improve retention. **Methods:** A retrospective analysis was conducted using data from the Treatment Site Information System (TSIS). A total of 40,272 PLHIV were identified, with exclusions for duplicates, transfers, non-regional health authority attendees, and deaths, leaving 12,542 for analysis. Retention was defined as active HIV care engagement without missed appointments >90 days, while loss to follow-up (LTFU) was >90 days without a clinic visit. Retention status was compared across demographic, socioeconomic, and clinical variables using descriptive statistics and logistic regression to identify factors significantly associated with retention. **Results:** Overall, 83% (n=10,466) were retained and 17% (n=2,076) were LTFU. Retention was significantly higher among females (84% vs. 82%, $p=0.003$) and married individuals (OR=2.0, $p<0.001$). Retention was lowest in PLHIV aged 20–44 years compared to those ≥ 50 ($p<0.001$). PLHIV receiving care outside their parish of residence were less likely to be retained (OR=0.7, $p<0.001$). Receiving SMS reminders was strongly associated with retention (OR=20.1, $p<0.001$). Clinical factors associated with higher retention included first CD4 <200 (OR=1.9, $p<0.001$) and viral load suppression, with undetectable VL increasing retention fivefold (OR=5.4, $p<0.001$). **Conclusion:** Retention in HIV care at public sites in Jamaica is generally high but suboptimal among younger age groups, males, and those accessing care outside their parish. Strengthening SMS reminder systems, targeted interventions for vulnerable groups, and expanded service access are recommended to improve retention.

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Prevalence of HIV Drug Resistance and Its Determinants Among Unsuppressed Persons Living with HIV in Jamaica

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Background: Jamaica has an estimated 28,000 persons living with HIV (PLHIV). The country has made steady progress in enrolling and retaining PLHIV in care on ART. While access to ART has increased over the last two decades, a growing global challenge has been the increasing prevalence of drug resistance. **Objective:** The study aims to support clinical management and advance progress toward the third UNAIDS 90-90-90 target—viral suppression by describing the profile of unsuppressed PLHIV in Jamaica with HIVDR mutations. **Methods:** A retrospective cross-sectional study of HIVDR samples of unsuppressed PLHIV in Jamaica was conducted between October 2024 to June 2025. A total of 212 plasma samples collected between 2018 and 2025 for HIV drug resistance (HIVDR) testing were reviewed. After clinical exclusions and laboratory quality control failures, 67 samples remained for analysis. We conducted statistical analysis using IBM SPSS version 20. Frequencies and proportions were used for categorical variables, Pearson's R test assessed associations between continuous variables, while the chi-square test examined associations

between categorical variables. **Results:** Of the 67 participants, 48% were male and 52% female, with a median age of 35 years. At ART initiation, 56% were classified at Stage III or IV (WHO staging). The median duration on ART at the time of HIVDR sample collection was 9 years (IQR: 7). HIVDR mutations were present in all participants. Most had NNRTI mutations (99%), followed by NRTI mutations (76%), PI mutations (28%), and INSTI mutations (12%). HIVDR mutations spanned multiple drug classes. The majority of participants demonstrated resistance to multiple drug classes: 43% (n=29) had resistance to two drug classes, 31% (n=21) to three drug classes, 22% (n=15) to a single drug class, and 3% (n=2) exhibited resistance across all four drug classes. There were no significant associations found between the number of HIVDR mutations and the following variables: viral load ($r(67) = 0.009$, $p = 0.94$), and time on ART ($r(66) = 0.1$, $p = 0.424$). However, a significant inverse correlation was observed between lower CD4 counts at ART initiation and the number of HIVDR mutations ($r(64) = -0.292$, $p = 0.019$). **Conclusion:** Findings from the study indicate a high prevalence of drug resistant mutations, particularly among the Nucleotide Reverse Transcriptase Inhibitors. This resistance will likely result in increased ART treatment failure and limited regimen options. It is imperative that the national HIV response implement more robust approaches to routine resistance testing, viral load monitoring, and other PLHIV support systems that will mitigate HIV drug resistance and maintain ART efficacy.

Mortality Patterns and Gaps in HIV Care: Insights from Jamaica, 2021–2023

Perry, D, Weller, CH, Brown, R, Gordon-Johnson, K, Douse-Mullings, A, Khan-Francis, R, Williams, A, Robb-Allen, A, Skyers, N, Beckford-Jarrett, S.

Background: In 2015, the World Health Organization recommended that PLHIV should be given Antiretroviral therapy immediately after diagnosis, which is aimed at reducing the mortality rate. Jamaica adopted the test-and-start approach in 2017. Despite these initiatives, PLHIV still experience challenges in treatment that can affect mortality. **Objective:** This study explores the demographic and clinical characteristics of persons diagnosed in Jamaica from 2021 to 2023, and how these factors influence mortality. **Methodology:** We analyzed data from PLHIV 15 years and older, who were diagnosed and enrolled at treatment sites in Jamaica between January 1, 2021, and December 31, 2023, to identify how demographic and clinical characteristics influence mortality. Chi-square tests, Kaplan-Meier Survival Analysis, and Cox Regression were used to examine survival patterns and factors associated with mortality during the study period. The data was extracted on July 4, 2025. **Results:** Of 3,056 individuals aged ≥ 15 years diagnosed between 2021 and 2023, 52% were male. Most were from the Southeast region, with the Northeast representing the smallest share. At baseline, 16.3% were WHO Stage 3, 18.9% Stage 4, and 10.9% had no CD4 recorded. Overall, 6.6% (n=201) died, 19.3% (n=590) were lost to follow-up, 4.5% (n=137) migrated, and 69.6% (n=2,128) remained alive and in care. Mortality was higher among males, those aged ≥ 60 years, persons with Stage 4 CD4 (<200), or missing baseline CD4, and those diagnosed in 2021. Independent predictors of death included age 25–59 years (AHR 4.72), ≥ 60 years (AHR 10.05), Stage 4 CD4 (AHR 2.38), and no CD4 recorded (AHR 8.81). **Conclusion:** Mortality outcomes have improved but may be underestimated due to loss to follow-up. Deaths were concentrated among older clients and those with advanced disease or missing CD4 data, underscoring gaps in testing, documentation, and retention. Strengthening early diagnosis, ART initiation, and CD4 monitoring remains critical to improving survival.

A Risk Assessment Matrix for Identifying Likelihood of Loss to Follow Up in Persons Living with HIV

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Background: Within the Southern Regional Health Authority (SRHA) in Jamaica, Lost to Follow-Up (LTFU) remains a persistent challenge that undermines continuity of HIV care. Continuous engagement in treatment supports adherence to antiretroviral therapy (ART), leading to improved viral suppression and reduced morbidity and mortality. As of June 2024, SRHA recorded 2,566 patients on register, of whom 2,074 (81%) were retained in care and 492 (19%) were classified as LTFU. **Methods:** To address this issue, a risk assessment matrix was developed to determine the likelihood of patients becoming LTFU by identifying and scoring predictors that negatively affect adherence. Each factor was assigned a weighted score (0–100), and an overall score was generated to categorize clients by risk level. Data were extracted from the National Treatment Site Information System (TSIS 2.0) for 2,399 eligible clients aged 18 years and older, who had been linked to SRHA treatment sites and on ART for at least five years. Statistical analyses, including Chi-Square, Phi, and Cramer's V tests, were conducted in SPSS to examine associations between predictors and LTFU outcomes. **Results:** The frequency analysis done on the data set found that the majority of clients who met the inclusion criteria were female (58.6%, n=1407) and that the highest percentage of clients were 50 years and older (36.8%, n=882). More than half the clients lived in rural areas (54.2%, n=1302) and almost a quarter of the sample had no address listed (23.3%, n=558). Majority (80%) of the eight (8) predictors contributed to the overall score statistical significance, $p < 0.001$. Patients with Low and Moderate risk scores showed very high retention (100% and 85% retained, respectively), while those in the Critical Risk category have the highest LTFU (61% lost). Gender and linkage time had no statistically significant differences. For the factors that had a Cramer's V result, it was found that the time spent in care (0.317), and the frequency of LTFU status (0.453) had higher association strengths than a client's age at registration (0.226) and their last viral load status (0.266). **Conclusion:** These results suggest that for better retention, lower risk scores were key. There were statistically significant differences in retention in care depending on factors such as age, address type, Frequency of loss to follow up, viral load status and length of time in care. Therefore, the higher the risk score, the greater the likelihood of becoming loss to follow-up.

30 Years of Advancement and Challenges in Early HIV Diagnosis in Jamaica – S01

Perry, D, Khan-Francis, R, Beckford Jarrett, S, Lawrence, J, Robb-Allen, A, McFarland, A, Skyers, N

Background: The first case of HIV in Jamaica was reported in 1982 and antiretrovirals (ARV) became available at public treatment sites in 2004. This study examines the epidemiology of HIV among persons living with HIV (PLHIV) from 1990 to 2020. **Methodology:** We analysed data from PLHIV first enrolled at HIV treatment sites in Jamaica between January 1990 to December 2020 to identify changes in PLHIV demographics, clinical stage, and location. The data span three periods: 1990-2003 (P1) pre-ARV access; 2004-2016 (P2) - Public ARV for PLHIV with CD4 <350; and 2017-2020 (P3) - Universal Test and Treat in Jamaica. We used chi-square and Kruskal-Wallis H tests to analyze changes across the periods. **Results:** 25,893 PLHIV registered at treatment sites: 731 were in P1, 19,479 in P2, and 5,683 in P3. Significant differences emerged in sex distribution ($\chi^2 (2) = 42.37, p < 0.001$), age ($\chi^2 (6) = 1142.42, p < 0.001$), area of residence ($\chi^2 (4) = 132.89, p < .001$), and WHO HIV stage at registration ($\chi^2 (6) = 368.70, p < 0.001$). Median age at registration increased from 29 in P1 to 36 in P3; ($H (2) = 260.59, p\text{-value} < 0.001$). Baseline CD4 counts increased from P1 to P3 ($H (2) = 424.26; p < .001$). Stage 4 diagnoses decreased in PLHIV ages under 15, 15-24, and 25-49, but not among PLHIV 50 years and older. Similarly, Stage 1 diagnosis increased across all age groups, except those in the 50 and over cohort. PLHIV residing in tourist areas increased while those

from urban areas decreased ($\chi^2(4) = 132.89, p < .001$). **Conclusion:** The decline in pediatric HIV cases and overall increase in early-stage diagnoses are notable public health achievements. However, Jamaica has an aging cohort of PLHIV entering treatment and targeted testing strategies and behavioural research are needed to reduce HIV transmission and late-stage diagnoses in older adults.

Factors Associated with Treatment Interruption – S02

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Objective: To determine the factors associated with recurring treatment interruption for PLHIV in Jamaica. **Methods:** A cross-sectional study with 3869 PLHIV retained in care in September 2023. The sample was divided into two cohorts: 1921 PLHIV, which had recurring treatment interruptions, and 1948 PLHIV, which had no treatment interruption. The chi-square test and multinomial logistic regression identified factors associated with having recurring treatment interruptions. **Results:** Among 3,869 PLHIV retained in care in September 2023, 55.1% were female and 44.9% males, with a median age of 44. Recurring treatment interruption was higher for PLHIV who registered for HIV care 0-14 years (79.9%), who accessed HIV treatment in NERHA (61.2%), who have been in care for 3 to 5 years (61.7%), and those who have an average appointment duration of 0-2 months (61.6%). The multivariable analysis showed recurring treatment interruption was likely for HIV treatment initiation at a younger age between 0 to 14 years (aOR=7.86 [4.46 – 13.72] vs. 60 and over), being in care for 6 to 8 years (aOR=1.66 [1.38 – 2.04] vs. greater than nine years), having an average appointment duration of 0 to 2 months (aOR=37.93 [10.04 – 143.36] vs. greater than eight months) and PLHIV accessing care in NERHA (aOR= 1.46 [1.12 – 1.91] vs WRHA). **Conclusions:** The study identified patient demographic characteristics, rigid programmatic guidelines, patient needs, behaviors, and preferences fuelling recurring interruption and disengagement in care. Efforts to reduce treatment interruptions should focus on addressing the challenges faced by patients who registered for HIV care at younger ages, improving access to consistent and flexible appointments, ensuring long-term viral suppression, and the need for region-specific interventions to address disengagement in care.

Factors Associated with Very Late Presentation Among PLHIV in Southern Jamaica – S02

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Background: Despite programmatic efforts, advanced HIV continues to be a challenge within the Caribbean region which increases the risk of mortality, and onward transmission. **Methods:** A retrospective cohort study was carried out on persons living with HIV linked to health facilities in Southern Jamaica during 2017 to 2022 to explore factors associated with stage of HIV diagnosis. Patients were classified into four categories based on the WHO guidelines for staging of CD4. Descriptive statistics and chi-square tests analyzed factors associated with CD4 stage. **Results:** Of the 410 patients, 35% were “very early” (CD4 $\geq$ 500 cells/ μ l), 20% were “fair early” (CD4 350-499 cells/ μ l), 23% were “late” (CD4 200 - 349 cells/ μ l) and 22% were “very late” (CD4 < 200 cells/ μ l). The study revealed that age was associated with stages of CD4 ($p < 0.01$). 7% aged 15-24yrs were diagnosed “very late”, compared to 23% aged 25-34yrs, 26% aged 35-49yrs and 28% aged 50 and over. Gender was also associated with stage of CD4 ($p < 0.01$). Males were more likely to be diagnosed very late (32%) compared to females (15%). Persons considered “alive” 20% were diagnosed “very late”, compared to 33% who migrated and 50 % considered deceased. **Conclusion:**

Factors such as gender, age at diagnosis, and patients now deceased were associated with very late presentation. Strengthening HIV testing targeting men and conversations on sexual health with older adults during clinic visits can reduce very late presentation.

Correcting the 2nd and 3rd 90s through Laboratory Surveillance – S01

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Background: Globally healthcare systems continue to experience significant impact by the Human Immunodeficiency Virus (HIV) epidemic. While Jamaica has made notable progress in achieving the first 90 of the UNAIDS 90-90-90 targets, challenges persist in achieving the second and third 90s, specifically in retaining persons living with HIV (PLHIV) in care and achieving viral suppression. Loss to follow-up (LTFU) remains a critical barrier, with significant gaps in national treatment cascades. Laboratory surveillance offers a potential solution to identify and track PLHIV categorized as LTFU by definition, but are accessing care at private and civil society organizations and not being documented as such in the Treatment Services Information System (TSIS). This allows for more accurate estimations of care engagement and viral suppression. **Methods:** A retrospective cohort study was conducted in March 2024 using data from the TSIS and District DISA Lab, an information management system utilized by the National Public Health Laboratory (NPHL). A total of 19,391 PLHIV linked to care in TSIS2 were analyzed, of which 3,315 were classified as LTFU. Data on viral load testing and suppression were manually compared across various healthcare providers, and statistical analyses were performed to assess trends in care engagement and viral suppression. **Results:** The study revealed that 54% of the LTFU cohort were female, and the age group 30-39 years accounted for the highest proportion (29%). Approximately 4% of the LTFU cohort had valid viral load results, with 70% achieving viral suppression. Public healthcare providers accounted for 92% of LTFU cases, while NGOs and private providers contributed 6% and 2%, respectively. Viral suppression rates among those with valid viral loads were highest in older age groups, with an overall suppression rate of 73%. **Conclusion:** Significant gaps in the care of PLHIV categorized as LTFU was identified through laboratory surveillance. Despite being classified as out of care, many PLHIV had valid viral load results and achieved viral suppression. Addressing LTFU through enhanced surveillance could lead to improved estimates of viral suppression and targeted interventions, supporting Jamaica's progress toward the UNAIDS 95-95-95 targets.